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ROLL NO SU91-BSRAM-F25-040

Lab 1

Lab task

TASK 1:

:

```cpp
#include <iostream>
using namespace std;
int main()
{
    char str[100];
    int length = 0;
    cout << "Enter a string: ";
    cin.getline(str, 100);
    char *p = str;
    while(*p != '\0')
    {
        length++;
        p++;
    }
    cout << "Reversed string: ";
    char *end = str + length - 1;
    while(end >= str)
    {
        cout << *end;
        end--;
    }
    return 0;
}
```

C:\Users\khalid\Downloads\1.exe

```
Enter a string: 55
Reversed string: 55
--------------------------------
Process exited after 2.903 seconds with return value 0
Press any key to continue . . .
```

Task 2

```
#include <iostream>
using namespace std;
int main() {
    int arr[5] = {10, 20, 30, 40, 50};
    int *ptr;
    ptr = &arr[2];
    printf("Value of 3rd element : %d\n", *ptr);
    printf("Address of 3rd element: %p\n", ptr);
    return 0;
}
```

Output

```
C:\Users\khalid\Downloads\1.exe
5 10 15 20 25 30
--------------------------------
Process exited after 1.035 seconds with return value 0
Press any key to continue . . .
```

Task 3

```
#include <iostream>
using namespace std;
int main() {
    int a, b;
    int *ptrA, *ptrB;
    printf("Enter value of a: ");
    scanf("%d", &a);
    printf("Enter value of b: ");
    scanf("%d", &b);
    ptrA = &a;
    ptrB = &b;
    printf("Value of a using pointer: %d\n", *ptrA);
    printf("Value of b using pointer: %d\n", *ptrB);
    return 0;
}
```

Output

```
C:\Users\khalid\Downloads\1.exe
Enter value of a: 15
Enter value of b: 19
Value of a using pointer: 15
Value of b using pointer: 19

--------------------------------
Process exited after 5.138 seconds with return value 0
Press any key to continue . . .
```

Task 4

```
#include <iostream>
using namespace std;
int main() {
    int a, b;
    int *ptrA, *ptrB;
    printf("Enter value of a: ");
    scanf("%d", &a);
    printf("Enter value of b: ");
    scanf("%d", &b);
    ptrA = &a;
    ptrB = &b;
    printf("Value of a using pointer: %d\n", *ptrA);
    printf("Value of b using pointer: %d\n", *ptrB);
    return 0;
}
```

```
C:\Users\khalid\Downloads\1.exe
Enter value of a: 15
Enter value of b: 19
Value of a using pointer: 15
Value of b using pointer: 19

--------------------------------
Process exited after 5.138 seconds with return value 0
Press any key to continue . . .
```

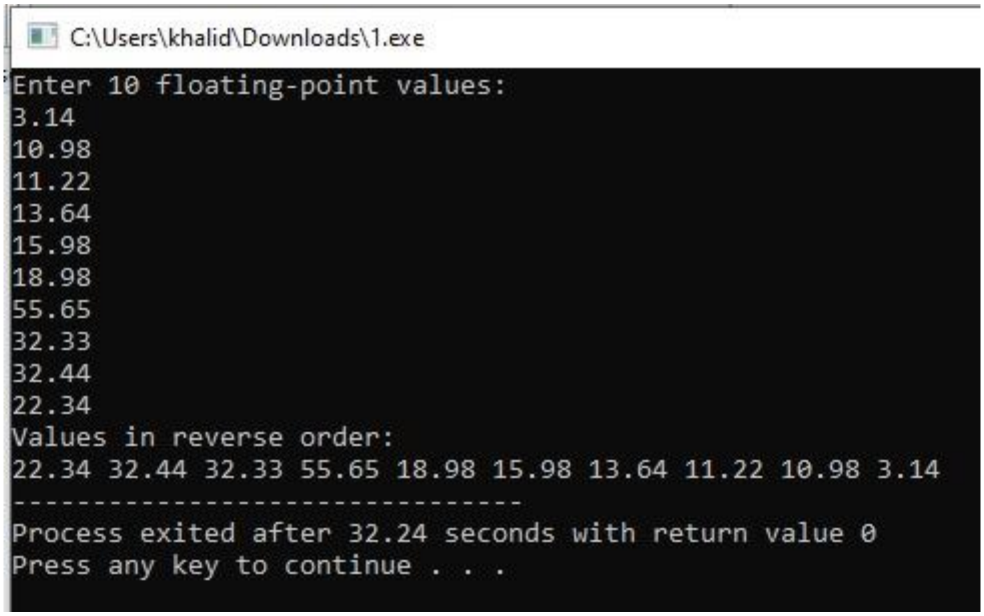
Task 5

```
include <iostream>
using namespace std;
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    int *ptr;
    ptr = arr;
    for(int i = 0; i < 5; i++) {
        printf("Address of element %d: %p\n", i + 1, ptr);
        ptr++;
    }
    return 0;
}
```

```
C:\Users\khalid\Downloads\1.exe
Address of element 1: 00000000006ffe00
Address of element 2: 00000000006ffe04
Address of element 3: 00000000006ffe08
Address of element 4: 00000000006ffe0c
Address of element 5: 00000000006ffe10

--------------------------------
Process exited after 1.059 seconds with return value 0
Press any key to continue . . .
```

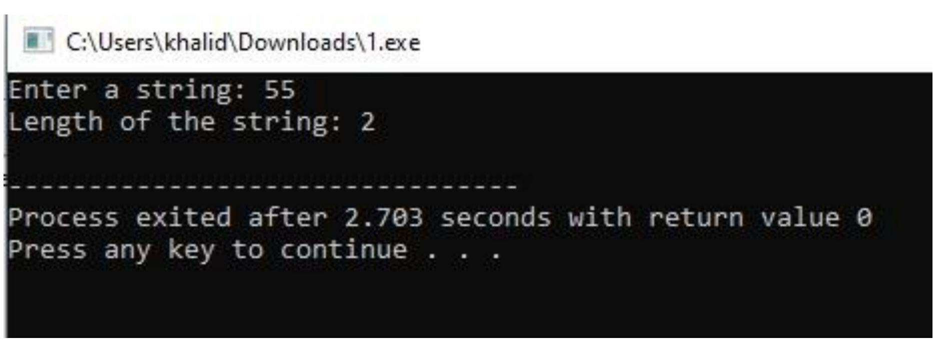
Task 6



C:\Users\khalid\Downloads\1.exe

```
Enter 10 floating-point values:
3.14
10.98
11.22
13.64
15.98
18.98
55.65
32.33
32.44
22.34
Values in reverse order:
22.34 32.44 32.33 55.65 18.98 15.98 13.64 11.22 10.98 3.14
--------------------------------
Process exited after 32.24 seconds with return value 0
Press any key to continue . . .
```

Task 7



C:\Users\khalid\Downloads\1.exe

```
Enter a string: 55
Length of the string: 2

--------------------------------
Process exited after 2.703 seconds with return value 0
Press any key to continue . . .
```

```
#include <iostream>
using namespace std;
int main() {
    char str[100];
    char *ptr;
    int length = 0;
    cout << "Enter a string: ";
    cin.getline(str, 100);
    ptr = str;
    while(*ptr != '\0') {
        length++;
        ptr++;
    }
    cout << "Length of the string: " << length << endl;
    return 0;
}
```

Home task

Task 1

```cpp
#include <iostream>
using namespace std;
int main() {
    int arr[100], n;
    int *ptr, *maxPtr;
    cout << "Enter number of elements (at least 8): ";
    cin >> n;
    if(n < 8) {
        cout << "Please enter at least 8 elements.\n";
        return 0;
    }
    cout << "Enter " << n << " integers:\n";
    for(int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    ptr = arr;
    maxPtr = arr;
    for(int i = 0; i < n; i++) {
        if(*ptr > *maxPtr) {
            maxPtr = ptr;
        }
        ptr++;
    }
    cout << "Maximum value: " << *maxPtr << endl;
    cout << "Address of max value: " << maxPtr << endl;
    return 0;
}
```

C:\Users\khalid\Downloads\1.exe

```
Enter number of elements (at least 8): 8
Enter 8 integers:
1
2
3
4
5
6
7
8
Maximum value: 8
Address of max value: 0x6ffc7c

--------------------------------
Process exited after 14.65 seconds with return value 0
Press any key to continue . . .
```

task 2

```cpp
#include <iostream>
using namespace std;
int main()
{
    int a, b, c, d, e;
    int* ptr[5];
    ptr[0] = &a;
    ptr[1] = &b;
    ptr[2] = &c;
    ptr[3] = &d;
    ptr[4] = &e;
    cout << "Enter five integer values: ";
    cin >> a >> b >> c >> d >> e;
    cout << "\nValues using pointer array:\n";
    for(int i = 0; i < 5; i++)
    {
        cout << "Value " << i+1 << " = " << *ptr[i] << endl;
    }
    cout << "\nAddresses stored in pointer array:\n";
    for(int i = 0; i < 5; i++)
    {
        cout << "Address " << i+1 << " = " << ptr[i] << endl;
    }
    return 0;
}
```

C:\Users\khalid\Downloads\1.exe

```
Enter five integer values: 5
4
3
2
1

Values using pointer array:
Value 1 = 5
Value 2 = 4
Value 3 = 3
Value 4 = 2
Value 5 = 1

Addresses stored in pointer array:
Address 1 = 0x6ffe04
Address 2 = 0x6ffe00
Address 3 = 0x6ffdfc
Address 4 = 0x6ffdf8
Address 5 = 0x6ffdf4

--------------------------------
Process exited after 8.058 seconds with return value 0
Press any key to continue . . .
```